

Problem Set 9

Integration by Substitution

September 25, 2026

Problem 1

(e)

$$\begin{aligned}u &= x^2 + 1 \\du &= 2x \, dx \\ \int \frac{6x}{x^2 + 1} \, dx &= \boxed{\int \frac{3}{u} \, du}\end{aligned}$$

(f)

$$\begin{aligned}u &= \ln t \\du &= \frac{1}{t} \, dt \\ \int \frac{\sqrt{\ln t}}{t} \, dt &= \boxed{\int \sqrt{u} \, du}\end{aligned}$$

(g)

$$\begin{aligned}u &= x^2 + 4x \\du &= 2x + 4 \, dx \\ \int (x + 2) \tan(x^2 + 4x) \, dx &= \boxed{\frac{1}{2} \int \tan u \, du}\end{aligned}$$

(h)

$$\begin{aligned}u &= \arctan x \\du &= \frac{1}{x^2 + 1} \, dx \\ \int \frac{(\arctan x)^2}{x^2 + 1} \, dx &= \boxed{\int u^2 \, du}\end{aligned}$$

Problem 2

(e)

$$u = 2x + 1$$

$$\begin{aligned}
 du &= 2 \, dx \\
 \frac{1}{2} du &= dx \\
 \int_0^{13} \frac{1}{\sqrt[3]{(1+2x)^2}} \, dx \\
 &= \frac{1}{2} \int_1^{27} \frac{1}{u^{\frac{2}{3}}} \, du \\
 &= \frac{3}{2} \left(u^{\frac{1}{3}} \right) \Big|_1^{27} \\
 &= \boxed{3}
 \end{aligned}$$

(f)

$$\begin{aligned}
 u &= x^2 + a^2 \\
 du &= 2x \, dx \\
 \frac{1}{2} du &= x \, dx \\
 \int_0^a x \sqrt{x^2 + a^2} \, dx \\
 &= \frac{1}{2} \int_{a^2}^{2a^2} \sqrt{u} \, du \\
 &= \frac{1}{3} (2\sqrt{2}a^3 - a^3) \\
 &= \boxed{\frac{a^3(2\sqrt{2} - 1)}{3}}
 \end{aligned}$$

(g)

$$\begin{aligned}
 u &= x - 1 \\
 du &= dx \\
 \int_1^2 x \sqrt{x-1} \, dx \\
 &= \int_0^1 (u+1) \sqrt{u} \, du \\
 &= \int_0^1 u^{\frac{3}{2}} + u^{\frac{1}{2}} \, du \\
 &= \left(\frac{2}{5} u^{\frac{5}{2}} + \frac{2}{3} u^{\frac{3}{2}} \right) \Big|_0^1 \\
 &= \boxed{\frac{16}{15}}
 \end{aligned}$$

(h)

$$u = \frac{1}{x^2}$$

$$\begin{aligned}
 du &= -\frac{2}{x^3} dx \\
 -\frac{1}{2} du &= \frac{dx}{x^3} \\
 \int_{\frac{1}{2}}^1 \frac{\cos(x^{-2})}{x^3} dx & \\
 &= \frac{1}{2} \int_1^4 \cos u \, du \\
 &= \boxed{\frac{\sin 4 - \sin 1}{2}}
 \end{aligned}$$

Problem 5

(a)

$$\begin{aligned}
 f(x) &= 2x \cos x^2 \\
 \cos x^2 = 0 &\implies x_k^2 = \frac{\pi}{2} + k\pi \\
 x_k &= \sqrt{\frac{(2k+1)\pi}{2}} \\
 \lim_{k \rightarrow \infty} x_{k+1} - x_k &= 0
 \end{aligned}$$

(b)

$$\begin{aligned}
 u &= x^2 \\
 du &= 2x \, dx \\
 \int 2x \cos x^2 \, dx &= \int \cos u \, du \\
 \int_0^{\sqrt{\frac{\pi}{2}}} 2x \cos x^2 \, dx &= \int_0^{\frac{\pi}{2}} \cos u \, du = 1 \\
 \int_{x_k}^{x_{k+1}} |2x \cos x^2| \, dx &= \int_{\frac{(2k+1)\pi}{2}}^{\frac{(2k+3)\pi}{2}} |\cos u| \, du = \boxed{2}
 \end{aligned}$$

The area of every full lobe is 2.

(c) The lobes of the original function are all stretched to the lobes of the standard function $\cos u$, which is always 2.

(d)

$$\int_0^{x_{2k}} 2x \cos x^2 \, dx = \boxed{1}$$

Problem 6

(a)

$$u = -x$$

$$du = -dx$$

$$\int_{-a}^0 f(x) dx = - \int_{-a}^0 f(-u) du = \int_0^a f(-u) du$$

(b)

$$\int_{-a}^a f(x) dx = \int_0^a f(x) dx + \int_0^a f(-x) dx = 2 \int_0^a f(x) dx$$

$$\int_{-a}^a f(x) dx = \int_0^a f(x) dx + \int_0^a f(-x) dx = 0$$

(c)

$$\int_{-2}^2 x^3 \cos x^2 dx = 0$$

$$\int_{-\pi/2}^{\pi/2} (x^2 \sin x + \cos x) dx = 0 + \int_{-\pi/2}^{\pi/2} \cos x dx = 2$$

(d) No, as the original function is not continuous on these intervals.

Problem 7

(a)

$$u = \sin x$$

$$du = \cos x dx$$

$$\int \sin x \cos x dx = \int u dx = \boxed{\frac{\sin^2 x}{2} + C}$$

(b)

$$u = \cos x$$

$$du = -\sin x dx$$

$$\int \sin x \cos x dx = - \int u dx = \boxed{-\frac{\cos^2 x}{2} + C}$$

(c)

$$\int \sin x \cos x dx = \frac{1}{2} \int \sin 2x dx$$

$$u = 2x$$

$$du = 2dx$$

$$\frac{1}{2} \int \sin 2x dx = \frac{1}{4} \int \sin u du = \boxed{-\frac{\cos 2x}{4} + C}$$

(d) As they merely differ by a constant, all three are correct anti-derivatives.

Problem 9

(e) No possible u -sub exists as the result of integration changes.

(f)

$$\begin{aligned} u &= \sin x \\ du &= \cos x \, dx \\ \int_0^\pi e^{\sin x} \cos x \, dx &= \int_0^0 e^u \, du = 0 \end{aligned}$$

The integral evaluates to zero as the function is symmetric about $x = \frac{\pi}{2}$.

Problem 10

(a)

$$\begin{aligned} T &= \frac{1}{60} \\ V_{avg} &= 60 \int_0^{\frac{1}{60}} 170 \sin(120\pi t) \, dt = 0 \end{aligned}$$

(b)

$$\begin{aligned} V_{avg} &= 170 \cdot 120 \int_0^{\frac{1}{120}} \sin(120\pi t) \, dt = 170 \cdot 120 \left(-\frac{\cos 120\pi t}{120\pi} \right) \Big|_0^{\frac{1}{120}} = \boxed{\frac{340}{\pi}} \\ \frac{340}{\pi} &\approx \boxed{108.23} \end{aligned}$$

(c)

$$\begin{aligned} v^2(t) &= 170^2 \sin^2(120\pi t) = \frac{170^2}{2} (1 - \cos 240\pi t) \\ v_{avg}^2 &= 60 \cdot \frac{170^2}{2} \int_0^{\frac{1}{60}} 1 - \cos 240\pi t \, dt = 30 \cdot 170^2 \left(t - \frac{\sin 240\pi t}{240} \right) \Big|_0^{\frac{1}{60}} = 14450 \\ V_{rms} &= \sqrt{14450} = \boxed{85\sqrt{2} \approx 120.2} \end{aligned}$$

(d)

$$\begin{aligned} T &= \frac{2\pi}{\omega} \\ V_{avg}^2 &= \frac{2\pi}{\omega} \int_0^{\frac{\omega}{2\pi}} A^2 \sin^2 \omega t \, dt \\ &= \frac{A^2 \omega}{4\pi} \int_0^{\frac{2\pi}{\omega}} 1 - \cos 2\omega t \, dt \end{aligned}$$

$$\begin{aligned}
&= \frac{A^2\omega}{4\pi} \left[t - \frac{\sin 2\omega t}{2\omega} \right] \Big|_0^{\frac{2\pi}{\omega}} \\
&= \frac{A^2\omega}{4\pi} \cdot \frac{2\pi}{\omega} = \frac{A^2}{2} \\
&\boxed{V_{rms} = \frac{A}{\sqrt{2}}}
\end{aligned}$$

(e) The label "120 V" refers to the answer to (c). By utilizing the V_{rms} , engineers may calculate values of the average power over certain intervals.

Problem 11

(a)

$$\begin{aligned}
u &= a - x \\
du &= -dx \\
\int_0^a f(x) dx &= \int_a^0 f(a - u) - du = \int_0^a f(a - u) du = \int_0^a f(a - x) dx
\end{aligned}$$

(b)

$$\begin{aligned}
I &= \int_0^{\frac{\pi}{2}} \frac{\sin^{2026} x}{\sin^{2026} x + \cos^{2026} x} dx \\
&= \int_0^{\frac{\pi}{2}} \frac{\sin^{2026}(\frac{\pi}{2} - x)}{\sin^{2026}(\frac{\pi}{2} - x) + \cos^{2026}(\frac{\pi}{2} - x)} dx = \int_0^{\frac{\pi}{2}} \frac{\cos^{2026} x}{\sin^{2026} x + \cos^{2026} x} dx \\
2I &= \int_0^{\frac{\pi}{2}} \frac{\sin^{2026} x + \cos^{2026} x}{\sin^{2026} x + \cos^{2026} x} dx = \int_0^{\frac{\pi}{2}} 1 dx = \frac{\pi}{2} \\
&\boxed{I = \frac{\pi}{4}}
\end{aligned}$$

(c)

$$\begin{aligned}
\int_0^\pi x g(\sin x) dx &= \int_0^\pi (\pi - x) g(\sin \pi - x) dx \\
&= \pi \int_0^\pi g(\sin x) dx - \int_0^\pi x g(\sin x) dx \\
2 \int_0^\pi x g(\sin x) dx &= \pi \int_0^\pi g(\sin x) dx \\
\int_0^\pi x g(\sin x) dx &= \frac{\pi}{2} \int_0^\pi g(\sin x) dx \\
\int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx &= \frac{\pi}{2} \int_0^\pi \frac{\sin x}{1 + \cos^2 x} dx \\
u &= \cos x \\
du &= -\sin x dx
\end{aligned}$$

$$\frac{\pi}{2} \int_0^\pi \frac{\sin x}{1 + \cos^2 x} dx = \frac{\pi}{2} \int_{-1}^1 \frac{1}{1 + u^2} du = \frac{\pi}{2} \arctan u \Big|_{-1}^1 = \boxed{\frac{\pi^2}{4}}$$

(d)

$$\begin{aligned}\tan u &= x \\ \sec^2 u \, du &= dx\end{aligned}$$

$$\begin{aligned}\int_0^1 \frac{\ln(1+x)}{1+x^2} dx &= \int_0^{\frac{\pi}{4}} \frac{\ln(1+\tan u)}{1+\tan^2 u} \cdot \sec^2 u \, du = \int_0^{\frac{\pi}{4}} \ln(1+\tan u) \, du \\ \int_0^{\frac{\pi}{4}} \ln(1+\tan u) \, du &= \int_0^{\frac{\pi}{4}} \ln(1+\tan(\frac{\pi}{4}-u)) \, du = \int_0^{\frac{\pi}{4}} \ln\left(\frac{2}{1+\tan u}\right) du \\ \int_0^{\frac{\pi}{4}} \ln\left(\frac{2}{1+\tan u}\right) du &= \int_0^{\frac{\pi}{4}} \ln 2 \, du - \int_0^{\frac{\pi}{4}} \ln(1+\tan u) \, du = \int_0^{\frac{\pi}{4}} \ln(1+\tan u) \, du \\ \int_0^{\frac{\pi}{4}} \ln(1+\tan u) \, du &= \frac{1}{2} \cdot \frac{\pi}{4} \cdot \ln 2 = \boxed{\frac{\pi}{8} \ln 2}\end{aligned}$$